

COMP 3069

Computer Graphics



Lecture 4:
Mathematics for Graphics
Autumn 2025

Contents

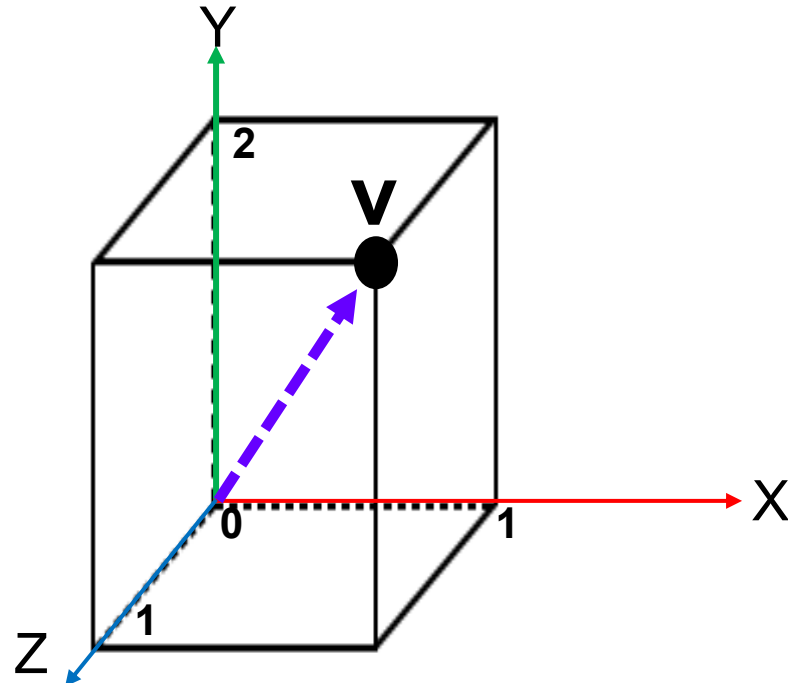
- ◆ Vectors and operations
- ◆ Normalising a vector
- ◆ Matrices and operations
- ◆ Multiplying a vector by a matrix
- ◆ Trigonometric functions $\cos()$ and $\sin()$ and their inverses
- ◆ Trigonometric functions $\cos()$ and $\sin()$ in a right triangle
- ◆ Pythagorean relation to the dot product



Vectors

- ◆ A vector $\mathbf{v}$ in 3-Dimensions is a triple. It can be written in row-vector form $\mathbf{v} = (v_0, v_1, v_2)$ of 3 elements or in column-vector form

$$\mathbf{v} = \begin{pmatrix} v_0 \\ v_1 \\ v_2 \end{pmatrix}$$



- ◆ The diagram above shows the 3-Dimensional Euclidean space with a vector $\mathbf{v} = (1, 2, 1)$

Vector Scale

向量的缩放，也就是向量的数乘

向量缩放是通过将向量中的所有元素乘以一个标量来实现的。

- ◆ Vector scaling is performed by multiplying all elements in the vector by a scalar

$$\mathbf{v}s = \begin{pmatrix} v_0 \cdot s \\ v_1 \cdot s \\ v_2 \cdot s \end{pmatrix}$$

向量 $\mathbf{v}$ 的每个分量都乘上同一个数字 s ，得到一个新的向量。

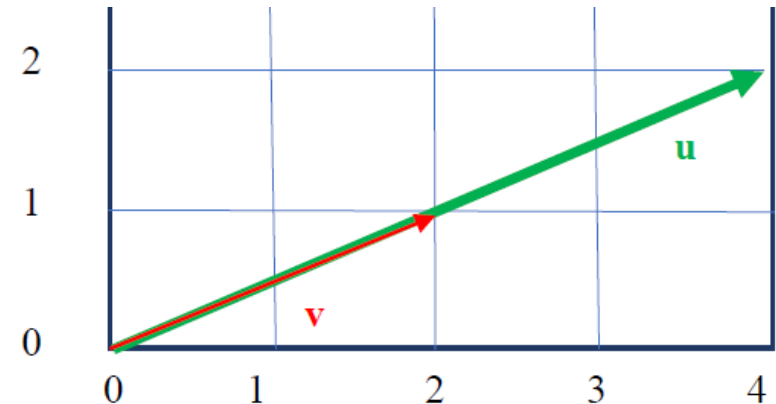
如果 $s > 1 \rightarrow$ 向量变长；

如果 $0 < s < 1 \rightarrow$ 向量变短；

如果 $s < 0 \rightarrow$ 向量方向反转。

- ◆ Scale $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ by 2, we have

$$\mathbf{u} = \mathbf{v}2 = \begin{pmatrix} 2 \cdot 2 \\ 1 \cdot 2 \\ 0 \cdot 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}$$



向量缩放就是“保持方向不变，改变长度”，通过把所有分量乘上同一个标量来实现。

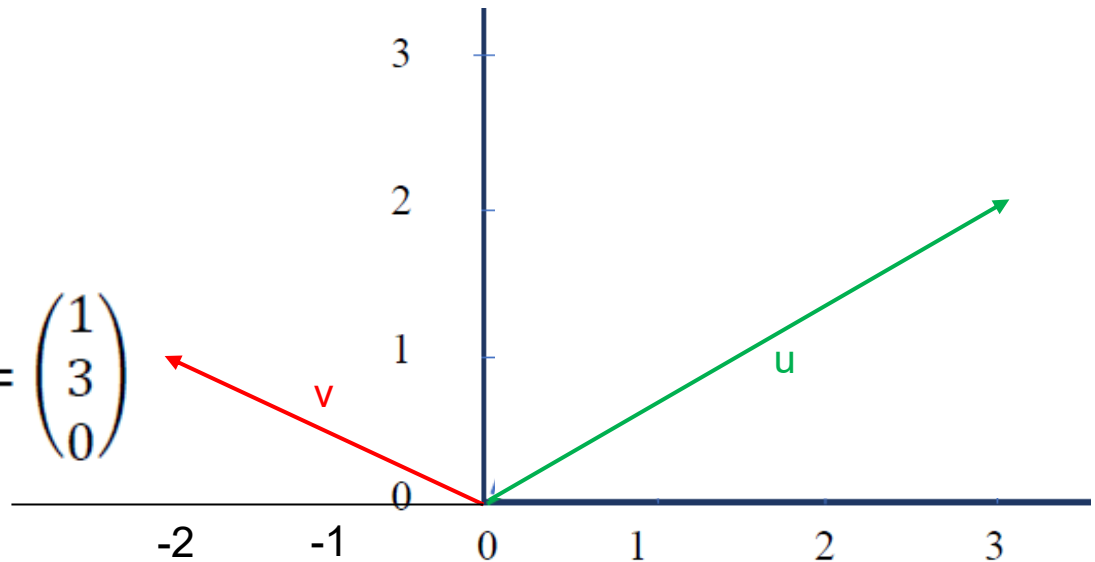
Vector Addition

- ◆ Vector addition of two vectors is performed on each element

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_0 \\ u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_0 \\ v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_0 + v_0 \\ u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

$$\mathbf{u} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 + -2 \\ 2 + 1 \\ 0 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$



Vector Addition

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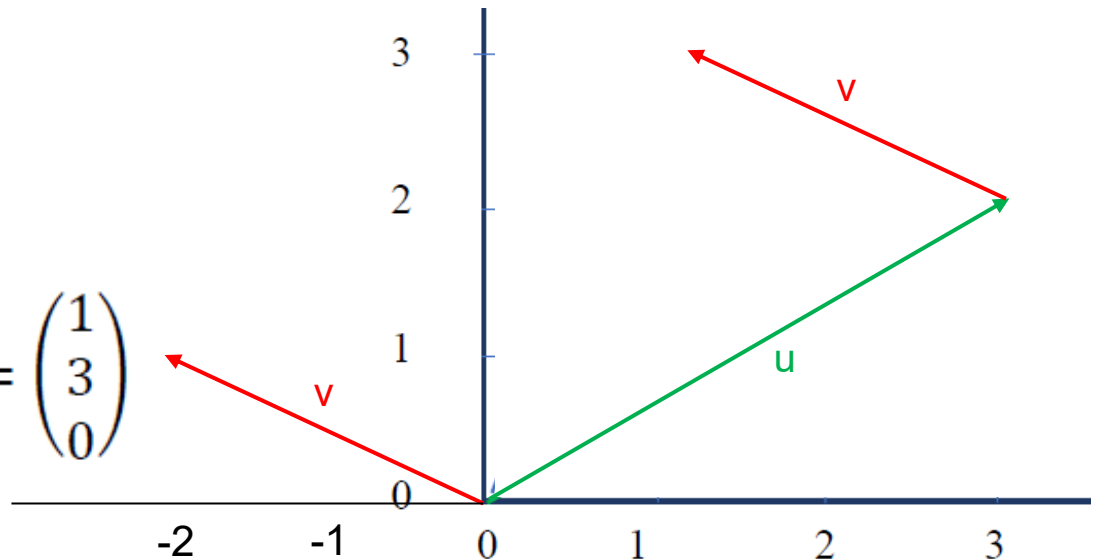
$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 + (-2) \\ 2 + 1 \\ 0 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$

在计算机图形学中，向量加法常用于：

- 位置变换：新位置 = 原位置 + 位移；
- 速度合成：总速度 = 向量叠加；
- 光照方向/摄像机移动：合并两个方向得到新方向。

例如：

$$\text{newPosition} = \text{oldPosition} + \text{velocity} \times \Delta t$$



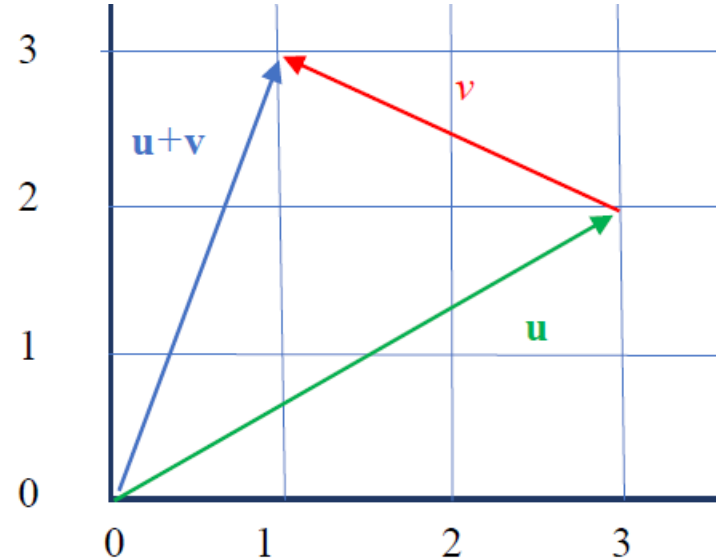
Vector Addition

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$$\mathbf{u} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 + (-2) \\ 2 + 1 \\ 0 + 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$



向量加法 = 对应分量相加。

几何上 = 把一个向量的“尾巴”接到另一个向量的“头上”，
结果向量从起点直达终点。

Vector Length

向量的长度（Vector Length），也称为向量的模（Magnitude）或范数（Norm）。

$$\|\mathbf{v}\| = \sqrt{v_0^2 + v_1^2 + v_2^2} \quad \|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

$$\mathbf{v} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \quad \|\mathbf{v}\| = \sqrt{3^2 + 2^2 + 0^2} = \sqrt{13} = 3.61$$

$$\sqrt{\begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}}$$



向量长度 ($\|\mathbf{v}\|$) 就是“它有多长”，
计算方法是各分量平方和的平方根，
或者等价地写作 $\sqrt{\mathbf{v} \cdot \mathbf{v}}$ 。

Vector Normalization

“向量归一化”，也常叫单位化 (Unit Vector Conversion)。
也就是如何把一个任意长度的向量“变成长度为 1 的向量”，同时保持方向不变。

- ◆ The notation for a normalised vector is $\hat{u}$, pronounced u-hat, which is a vector of length 1

• 符号“ $\hat{u}$ ” (读作 u-hat) 表示“单位向量” (unit vector)。

• 它是一个长度为 1 的向量，方向与原向量相同。

• 归一化的目标：保持方向不变，只改变长度，使长度 = 1。

$$\frac{1}{\|\mathbf{v}\|} \cdot \mathbf{v} \quad \mathbf{v} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix} \quad \|\mathbf{v}\| = 3.61$$

$$\hat{u} = \mathbf{v} \cdot \frac{1}{3.61} = \begin{pmatrix} \frac{3}{3.61} \\ \frac{2}{3.61} \\ \frac{0}{3.61} \end{pmatrix} = \begin{pmatrix} 0.83 \\ 0.55 \\ 0 \end{pmatrix}$$



| 向量归一化 = 用向量除以它的长度，得到长度为 1、方向不变的单位向量。

公式: $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$

Dot Product 点积

- ◆ In 3-dimensional space, dot product can be written as

$$\mathbf{u} \cdot \mathbf{v} = (u_0 * v_0) + (u_1 * v_1) + (u_2 * v_2) \quad \text{代数形式：就是把对应分量相乘后相加}$$

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta \quad \text{几何形式}$$

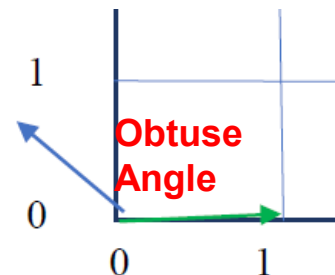
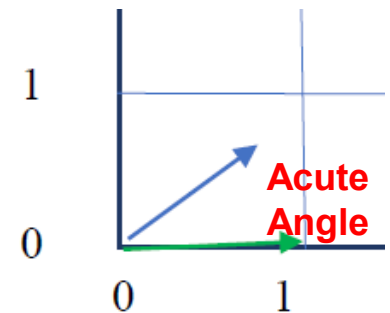
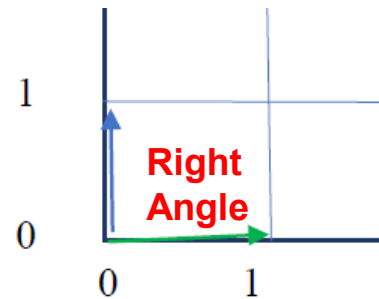
- ◆ Three insightful rules of dot product:

$$\mathbf{u} \cdot \mathbf{v} = 0 \Leftrightarrow \mathbf{u} \perp \mathbf{v}$$

$$\mathbf{u} \cdot \mathbf{v} > 0 \Leftrightarrow 0 \leq \theta < \frac{\pi}{2}$$

$$\mathbf{u} \cdot \mathbf{v} < 0 \Leftrightarrow \frac{\pi}{2} < \theta \leq \pi$$

- Right Angle (直角) $\rightarrow$ 点积 = 0
- Acute Angle (锐角) $\rightarrow$ 点积 > 0
- Obtuse Angle (钝角) $\rightarrow$ 点积 < 0

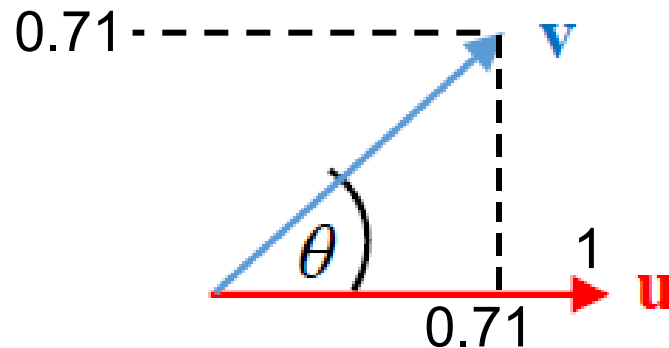


Dot Product of Unit Vectors

单位向量的点积

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 0.71 \\ 0.71 \\ 0 \end{pmatrix} \quad \|\mathbf{u}\| = 1 \quad \|\mathbf{v}\| = 1$$

$$\mathbf{u} \cdot \mathbf{v} = (1 * 0.71) + (0 * 0.71) + (0 * 0) = 0.71$$



Dot Product of Unit Vectors

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} 0.71 \\ 0.71 \\ 0 \end{pmatrix}$$

$$\|\mathbf{u}\| = \sqrt{1^2 + 0^2 + 0^2} = 1 \quad \|\mathbf{v}\| = \sqrt{0.71^2 + 0.71^2 + 0^2} = 1$$

$$\mathbf{u} \cdot \mathbf{v} = (1 * 0.71) + (0 * 0.71) + (0 * 0) = 0.71$$

$$\|\mathbf{u}\| \|\mathbf{v}\| \cos \theta = \mathbf{u} \cdot \mathbf{v}$$

$$1 \times 1 \times \cos \theta = 0.71$$

$$\cos \theta = 0.71$$

$$\theta = \arccos(0.71)$$

$$\approx 0.7813 \text{ rad}$$

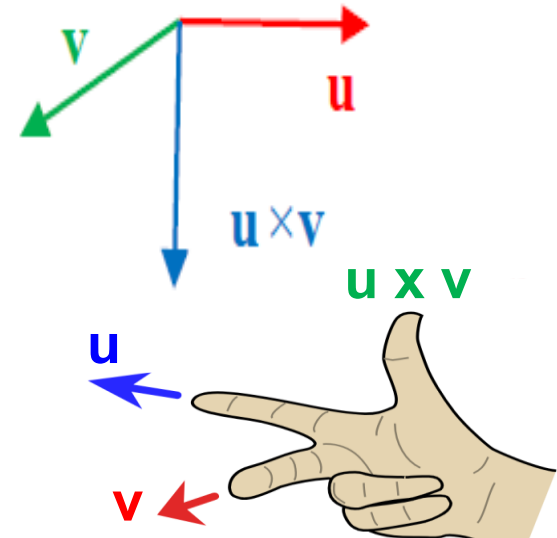
$$\approx 45^\circ$$

Cross Product 向量叉积

- The cross product of two 3-D vectors is defined as

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} (u_1 * v_2) - (u_2 * v_1) \\ (u_2 * v_0) - (u_0 * v_2) \\ (u_0 * v_1) - (u_1 * v_0) \end{pmatrix}$$

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



它的结果不是一个数，而是另一个向量。

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} (u_1 * v_2) - (u_2 * v_1) \\ (u_2 * v_0) - (u_0 * v_2) \\ (u_0 * v_1) - (u_1 * v_0) \end{pmatrix} = \begin{pmatrix} (0 * 1) - (0 * 0) \\ (0 * 0) - (1 * 1) \\ (1 * 0) - (0 * 0) \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

<https://www.mathsisfun.com/algebra/vectors-cross-product.html>

<https://www.khanacademy.org/math/multivariable-calculus/thinking-about-multivariable-function/x786f2022:vectors-and-matrices/a/cross-products-mvc>

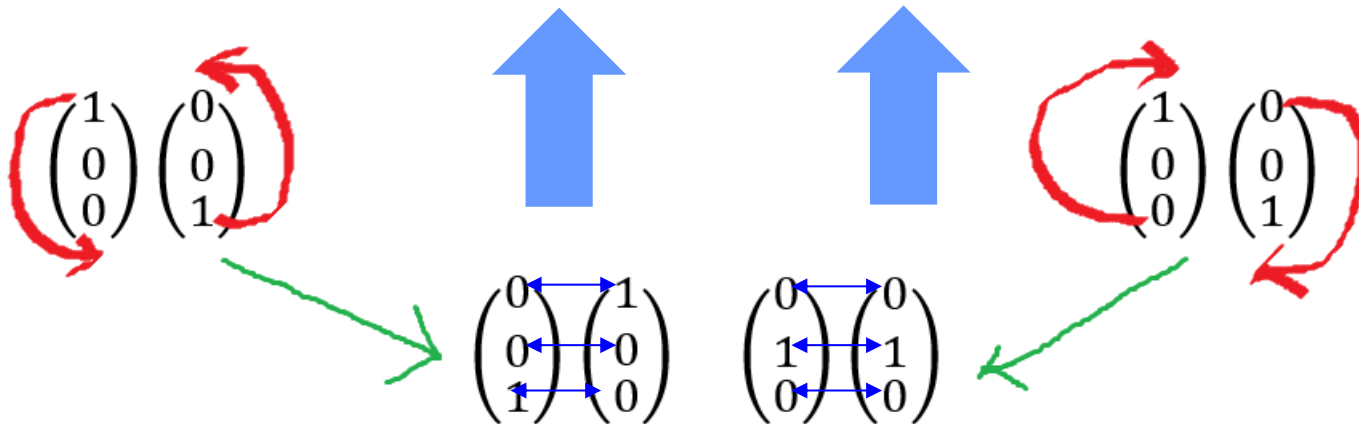


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Cross Product

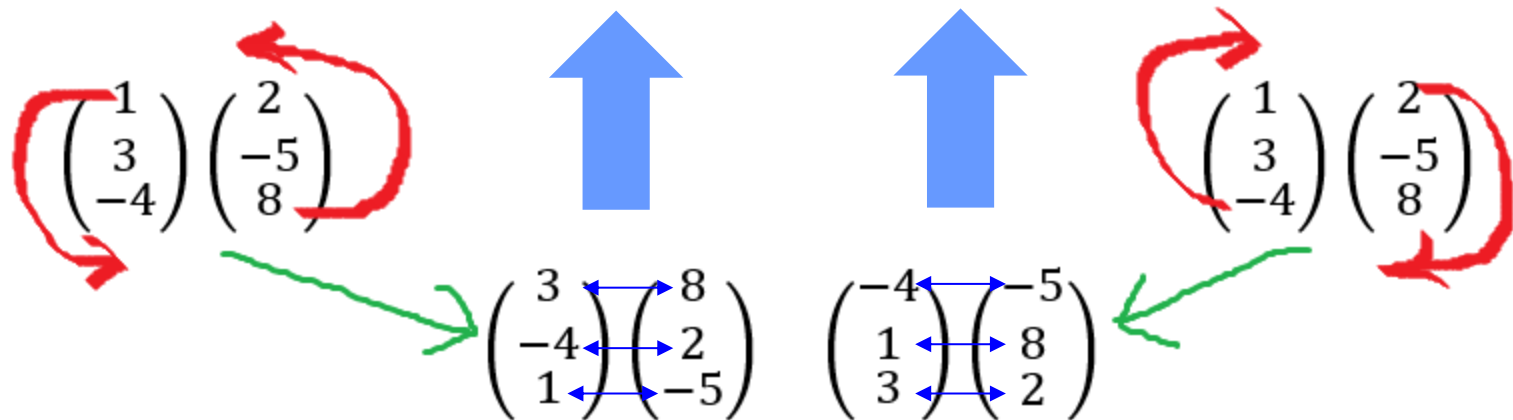
$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} (0)(1) & - & (0)(0) \\ (0)(0) & - & (1)(1) \\ (1)(0) & - & (0)(0) \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$



How to remember the steps?

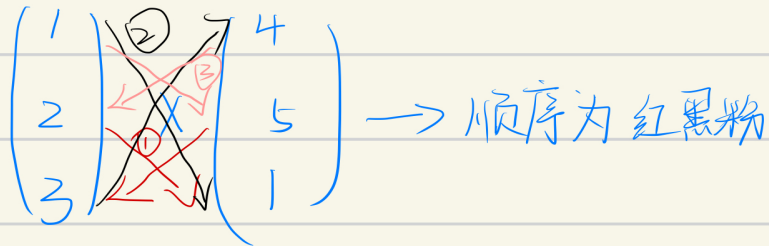
Cross Product

$$\begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} \times \begin{pmatrix} 2 \\ -5 \\ 8 \end{pmatrix} = \begin{pmatrix} (3)(8) - (-4)(-5) \\ (-4)(2) - (1)(8) \\ (1)(-5) - (3)(2) \end{pmatrix} = \begin{pmatrix} 4 \\ -16 \\ -11 \end{pmatrix}$$



How to remember the steps?

Cross Product



$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} (?) (?) - (?) (?) \\ (?) (?) - (?) (?) \\ (?) (?) - (?) (?) \end{pmatrix} = \begin{pmatrix} -13 \\ 11 \\ -3 \end{pmatrix}$$

(1) 第一行:

$$u_1 v_2 - u_2 v_1 = (2)(1) - (3)(5) = 2 - 15 = -13$$

(2) 第二行:

$$u_2 v_0 - u_0 v_2 = (3)(4) - (1)(1) = 12 - 1 = 11$$

(3) 第三行:

$$u_0 v_1 - u_1 v_0 = (1)(5) - (2)(4) = 5 - 8 = -3$$

Matrices 矩阵

- ◆ A matrix M is rectangular grid of $r \times c$ scalars, arranged in c columns and r rows. We mostly use 4×4 matrices in graphics.

$$M = \begin{pmatrix} m_{0,0} & m_{0,1} & \cdots & m_{0,c-1} \\ m_{1,0} & m_{1,1} & \cdots & m_{1,c-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{r-1,0} & m_{r-1,1} & \cdots & m_{r-1,c-1} \end{pmatrix}$$

- 一个矩阵 (Matrix) 是一个由数字 (scalars) 组成的长方形网格;
- 它有 r 行 (rows) 和 c 列 (columns);
- 在图形学 (graphics) 中, 最常用的是 4×4 矩阵 (尤其在三维变换中)。

- ◆ The identity matrix is 单位矩阵

$$I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- 主对角线 (从左上到右下) 的元素全是 1;
- 其他位置全是 0;
- 它的大小常是 4×4 (在 3D 图形变换中)。

Transpose of a Matrix 矩阵的转置

- During a transpose operation, the rows become the columns and the columns become rows 行变列，列变行

$$\mathbf{M} = \begin{pmatrix} m_{0,0} & m_{0,1} & \cdots & m_{0,c-1} \\ m_{1,0} & m_{1,1} & \cdots & m_{1,c-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{r-1,0} & m_{r-1,1} & \cdots & m_{r-1,c-1} \end{pmatrix}$$

$$\mathbf{M}^T = \begin{pmatrix} m_{0,0} & m_{1,0} & \cdots & m_{r-1,0} \\ m_{0,1} & m_{1,1} & \cdots & m_{r-1,1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{0,c-1} & m_{1,c-1} & \cdots & m_{r-1,c-1} \end{pmatrix}$$

$$\mathbf{M} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{M}^T = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$$

Matrix Addition 矩阵加法

◆ Two matrices M & N of equal size can be added

只有当两个矩阵的大小相同（即行数、列数完全一致）时，才能进行加法运算

$$\mathbf{M} = \begin{pmatrix} m_{0,0} & m_{0,1} & \cdots & m_{0,c-1} \\ m_{1,0} & m_{1,1} & \cdots & m_{1,c-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{r-1,0} & m_{r-1,1} & \cdots & m_{r-1,c-1} \end{pmatrix}$$

$$\mathbf{N} = \begin{pmatrix} n_{0,0} & n_{0,1} & \cdots & n_{0,c-1} \\ n_{1,0} & n_{1,1} & \cdots & n_{1,c-1} \\ \vdots & \vdots & \ddots & \vdots \\ n_{r-1,0} & n_{r-1,1} & \cdots & n_{r-1,c-1} \end{pmatrix}$$

$$\mathbf{M} + \mathbf{N} = \begin{pmatrix} m_{0,0} + n_{0,0} & m_{0,1} + n_{0,1} & \cdots & m_{0,c-1} + n_{0,c-1} \\ m_{1,0} + n_{1,0} & m_{1,1} + n_{1,1} & \cdots & m_{1,c-1} + n_{1,c-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{r-1,0} + n_{r-1,0} & m_{r-1,1} + n_{r-1,1} & \cdots & m_{r-1,c-1} + n_{r-1,c-1} \end{pmatrix}$$

$$\mathbf{M} = \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix} \quad \mathbf{N} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \quad \mathbf{M} + \mathbf{N} = \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0+2 & 4+3 \\ 1+0 & 2+1 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$$

Matrix Multiplication 矩阵乘法

- Two matrices **M** and **N** can be multiplied only if **M** is of size $p \times q$ and **N** is of size $q \times r$. The resulting matrix **T** is of size $p \times r$.

$$M = \begin{pmatrix} m_{0,0} & m_{0,1} & \cdots & m_{0,q-1} \\ m_{1,0} & m_{1,1} & \cdots & m_{1,q-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{p-1,0} & m_{p-1,1} & \cdots & m_{p-1,q-1} \end{pmatrix}$$

$$N = \begin{pmatrix} n_{0,0} & n_{0,1} & \cdots & n_{0,r-1} \\ n_{1,0} & n_{1,1} & \cdots & n_{1,r-1} \\ \vdots & \vdots & \ddots & \vdots \\ n_{q-1,0} & n_{q-1,1} & \cdots & n_{q-1,r-1} \end{pmatrix}$$

$$T = MN$$

$$= \begin{pmatrix} n_{0,0}m_{0,0} + n_{1,0}m_{0,1} + \cdots + n_{q-1,0}m_{0,q-1} & n_{0,1}m_{0,0} + n_{1,1}m_{0,1} + \cdots + n_{q-1,1}m_{0,q-1} & \cdots & n_{0,r-1}m_{0,0} + n_{1,r-1}m_{0,1} + \cdots + n_{q-1,r-1}m_{0,q-1} \\ n_{0,0}m_{1,0} + n_{1,0}m_{1,1} + \cdots + n_{q-1,0}m_{1,q-1} & n_{0,1}m_{1,0} + n_{1,1}m_{1,1} + \cdots + n_{q-1,1}m_{1,q-1} & \cdots & n_{0,r-1}m_{1,0} + n_{1,r-1}m_{1,1} + \cdots + n_{q-1,r-1}m_{1,q-1} \\ \vdots & \vdots & \ddots & \vdots \\ n_{0,0}m_{p-1,0} + n_{1,0}m_{p-1,1} + \cdots + n_{q-1,0}m_{p-1,q-1} & n_{0,1}m_{p-1,0} + n_{1,1}m_{p-1,1} + \cdots + n_{q-1,1}m_{p-1,q-1} & \cdots & n_{0,r-1}m_{p-1,0} + n_{1,r-1}m_{p-1,1} + \cdots + n_{q-1,r-1}m_{p-1,q-1} \end{pmatrix}$$

- 若矩阵 M 的大小为 $p \times q$, 即 M 有 p 行 q 列;
- 而矩阵 N 的大小为 $q \times r$, 即 N 有 q 行 r 列;
- 那么 $M \times N$ 是可乘的;
- 结果矩阵 $T = MN$ 的大小是 $p \times r$ 。

换句话说:

内维相等 (q), 外维决定结果 ($p \times r$)。

例如:

$$(2 \times 3) \times (3 \times 4) \Rightarrow (2 \times 4)$$



Example of Matrix Multiplication

$$\mathbf{M} = \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix}$$

$$\mathbf{N} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$$

$$\mathbf{T} = \mathbf{MN} = \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix} \times \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 \times 2 + 4 \times 0 & 0 \times 3 + 4 \times 1 \\ 1 \times 2 + 2 \times 0 & 1 \times 3 + 2 \times 1 \end{pmatrix} = \begin{pmatrix} 0 & 4 \\ 2 & 5 \end{pmatrix}$$

- ◆ Matrix-matrix multiplication is not commutative,
 $\mathbf{MN} \neq \mathbf{NM}$ 没有交换律

$$\mathbf{S} = \mathbf{NM} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 \times 2 + 1 \times 3 & 4 \times 2 + 2 \times 3 \\ 0 \times 0 + 1 \times 1 & 4 \times 0 + 2 \times 1 \end{pmatrix} = \begin{pmatrix} 3 & 14 \\ 1 & 2 \end{pmatrix}$$

$$\mathbf{T} = \mathbf{MN} = \begin{pmatrix} 0 & 4 \\ 2 & 5 \end{pmatrix} \neq \mathbf{S} = \mathbf{NM} = \begin{pmatrix} 3 & 14 \\ 1 & 2 \end{pmatrix}$$

Example of Matrix Multiplication

Are the following matrix multiplications possible?

$$\begin{bmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 5 \end{bmatrix} \times \begin{bmatrix} 1 & -1 \\ 0 & 2 \\ 1 & 5 \end{bmatrix} \quad \text{NO}$$

3×2 3×2

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 4 \\ 5 & 8 \end{bmatrix} \quad \text{YES}$$

2×3 3×2



Example of Matrix Multiplication

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 4 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix}$$

$$(1)(0) + (2)(1) + (4)(5) = 22$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 4 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} 22 & \square \\ 54 & \square \end{bmatrix}$$

$$(1)(1) + (2)(4) + (4)(8) = 41$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 4 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} 22 & \square \\ \square & \square \end{bmatrix}$$

$$(3)(0) + (9)(1) + (9)(5) = 54$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 9 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 4 \\ 5 & 8 \end{bmatrix} = \begin{bmatrix} 22 & 41 \\ 54 & \square \end{bmatrix}$$

$$(3)(1) + (9)(4) + (9)(8) = 111$$

$$\begin{bmatrix} 22 & 41 \\ 54 & 111 \end{bmatrix}$$



Matrix-Vector Multiplication

- ◆ If we have a matrix **M** of size $p \times q$, and a vector **v** of size q , then **M** and **v** can be multiplied

$$\mathbf{M} = \begin{pmatrix} m_{0,0} & m_{0,1} & \cdots & m_{0,q-1} \\ m_{1,0} & m_{1,1} & \cdots & m_{1,q-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{p-1,0} & m_{p-1,1} & \cdots & m_{p-1,q-1} \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{q-1} \end{pmatrix}$$

$$(p \times q) \cdot (q \times 1) = (p \times 1)$$

$$\mathbf{u} = \mathbf{M}\mathbf{v} = \begin{pmatrix} m_{0,0} & m_{0,1} & \cdots & m_{0,q-1} \\ m_{1,0} & m_{1,1} & \cdots & m_{1,q-1} \\ \vdots & \vdots & \ddots & \vdots \\ m_{p-1,0} & m_{p-1,1} & \cdots & m_{p-1,q-1} \end{pmatrix} \begin{pmatrix} v_0 \\ v_1 \\ \vdots \\ v_{q-1} \end{pmatrix} = \begin{pmatrix} v_0 m_{0,0} + v_1 m_{0,1} \cdots + v_{q-1} m_{0,q-1} \\ v_0 m_{1,0} + v_1 m_{1,1} \cdots + v_{q-1} m_{1,q-1} \\ \vdots \\ v_0 m_{p-1,0} + v_1 m_{p-1,1} \cdots + v_{q-1} m_{p-1,q-1} \end{pmatrix}$$

Example of Matrix-Vector Multiplication

- ◆ Vector **v** and each row of **M** have a dot product operation applied 矩阵的列数 (columns) 必须等于向量的行数 (rows)

$$\mathbf{M} = \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix}$$

$$\mathbf{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\mathbf{u} = \mathbf{M}\mathbf{v} = \begin{pmatrix} 0 & 4 \\ 1 & 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 * 3 + 4 * 2 \\ 1 * 3 + 2 * 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 7 \end{pmatrix}$$



Identity Matrix-Vector Multiplication

单位矩阵与向量相乘

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_q \end{pmatrix}$$

单位矩阵乘任何向量，结果不变。

也就是：

$$\mathbf{I} \times \mathbf{v} = \mathbf{v}$$

它在图形学中表示“不改变方向、不改变大小、不改变位置”的变换。

$$\mathbf{u} = \mathbf{I} \times \mathbf{v} = \begin{pmatrix} 1v_1 + 0v_2 \cdots + 0v_q \\ 0v_1 + 1v_2 \cdots + 0v_q \\ \vdots \\ 0v_1 + 0v_2 \cdots + 1v_q \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_q \end{pmatrix} = \mathbf{v}$$

Trigonometry 三角学

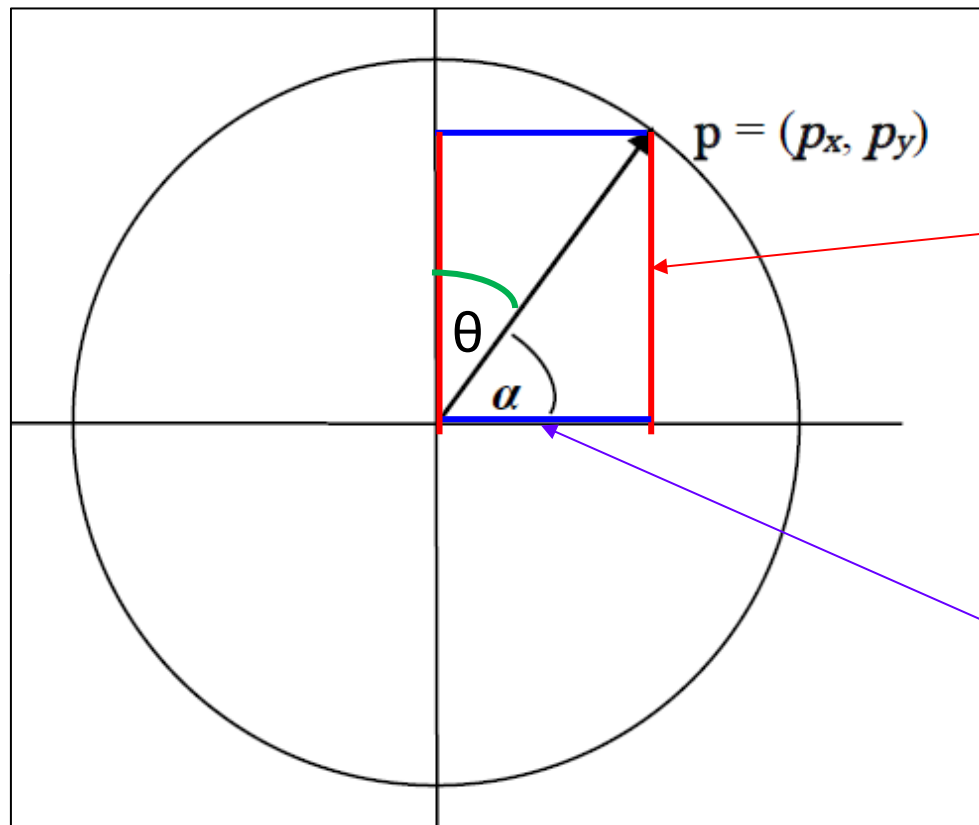
- ◆ If $\mathbf{p} = (p_x, p_y)$ is a unit vector, the trigonometric functions are $p_x = \cos \alpha$ and $p_y = \sin \alpha$

- 向量 $\mathbf{p}$ 是一个单位向量 (即长度 = 1);

- 它的方向与角度 α 有关;

- 那么它的分量可以写为:

$$p_x = \cos \alpha, \quad p_y = \sin \alpha$$

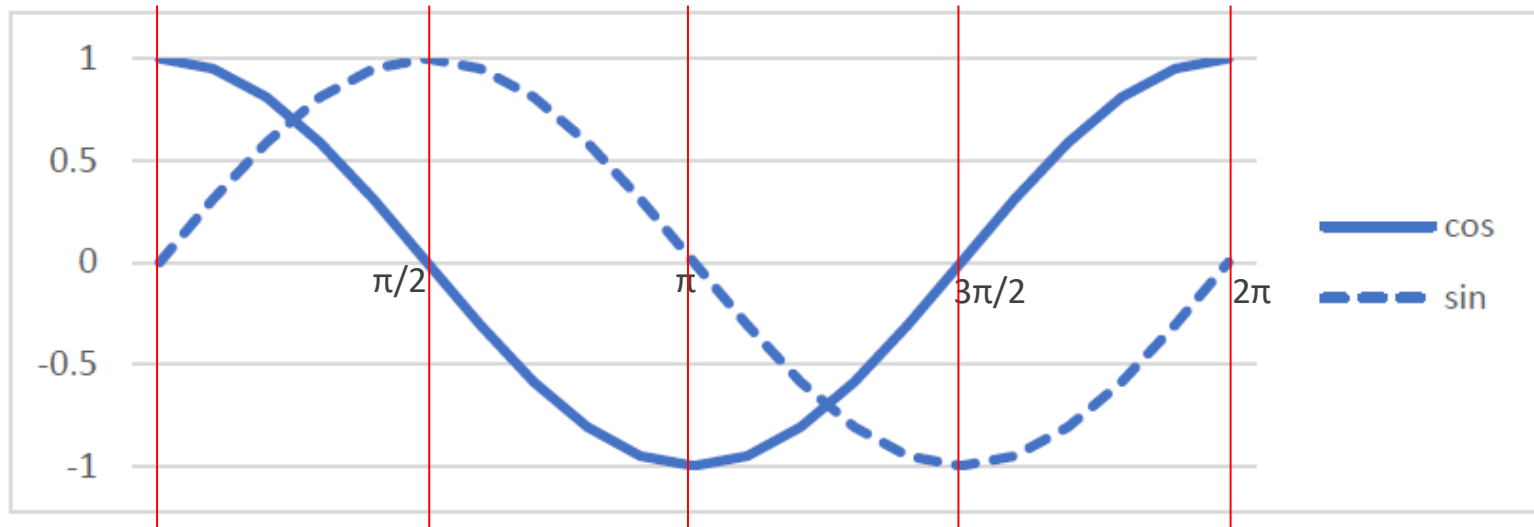


对边/斜边
 $p_y = \sin \alpha$
 $p_y = \cos \theta$

邻边/斜边
 $p_x = \cos \alpha$
 $p_x = \sin \theta$



sin() and cos()



- ◆ The inverses of the trigonometric functions are

$$p_x = \cos \alpha \Leftrightarrow \alpha = \arccos p_x$$

$$p_y = \sin \alpha \Leftrightarrow \alpha = \arcsin p_y$$

sin 和 cos 描述角度与坐标的关系,

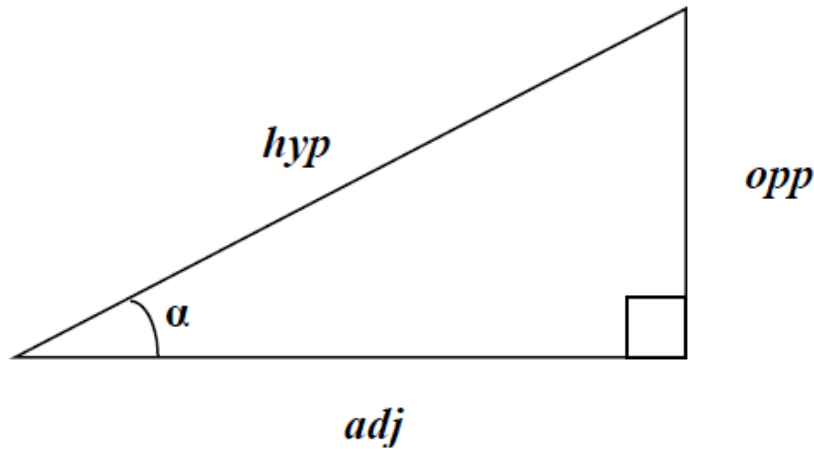
而 arcsin 与 arccos 则能从坐标反推出角度:

$$p_x = \cos \alpha \Rightarrow \alpha = \arccos(p_x), \quad p_y = \sin \alpha \Rightarrow \alpha = \arcsin(p_y)$$



Pythagorean Relation

勾股定理



$$\sin \alpha = \frac{opp}{hyp}$$

$$\cos \alpha = \frac{adj}{hyp}$$

- ◆ The Pythagorean relation is

$$hyp^2 = opp^2 + adj^2$$



References

- ◆ *COMP3069 Lecture notes: Chapter 4 - Graphics Programming, OpenGL & GLSL*
- ◆ *Interactive Computer Graphics: A Top-Down Approach with Shader-Based OpenGL, Chapter 3*